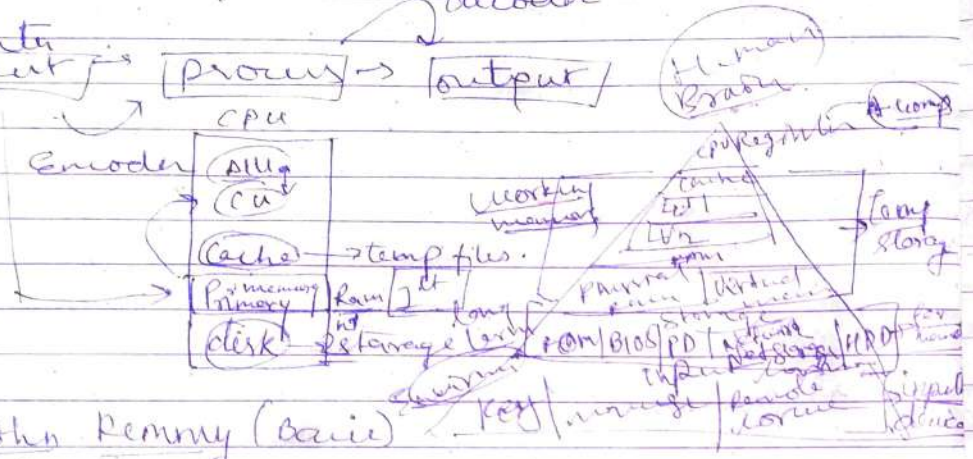
Women safety top most

→ Computer fundamental (Sarkar study)



John Kemmy (Basic)

	Processor			
I st (1946)	Valum	No OS	UNIVAC	John mauchly/JP Eckert
II nd	transistor ing tape	Cobol/Fortran	IBM 2030	
III rd	IC's	Pascal	IBM 360	
IV th	VLSI	90 OS	APPLE II	
V th	ULSI (micro)	INTERNET	PARAM	

upment

Nano | quantum

History

— 3000 ago | Abacus china

France

— Blaze pascal | 1645 | + - ×÷

— Jacquard's loom | 1801 |

America

Charles babbage | calci
— Lady Ada Augusta

Dr Howard Aiken's Mark I
& Grace | 12 m ÷

ABC | 1st Electronic Device
— Napier bone
— Slide Rule
— Leibnitz
— Curvus Calculator | Hyman | 1820

1970 Alan kay | thought
1979 William moggridge | laptop design
1983 Gavilan | 64 Kb MS-DOS | touch
CDAC → PARAM SERIES
Anurag → Mr. Seymour Cray | 1976

-9 → 6hr 9min | 2:03 pm

Classification of comp

- By Type
  - Analog (thermometer) (counting)
  - Digital (clock)
  - Hybrid (medical device)
- By size
  - micro (laptop)
  - mini (phone)
  - Mainframe
  - super
- By Purpose
  - General (Home)
  - Special (weather)

024 TB = 1 PB (Peta)
= 1 EB (Exa)
= 1 ZB (Zetta)
= 1 YB (Yotta)
= 1 (brontO)
= 1 (Geop)

Memory

- Primary
  - ROM (BIOS) (Readonly)
  - PROM
  - EPROM
  - EEPROM
  - Cache
  - RAM
    - SRAM
    - DRAM
    - DBR RAM
    - DDR2 RAM
    - DDR3
    - DDR4
- Secondary
  - HDD 7200 RPM 3.5
  - SSD 2.5
  - USB
  - GD (AR
  - CD
  - DVD
  - floop
  - Tape drive

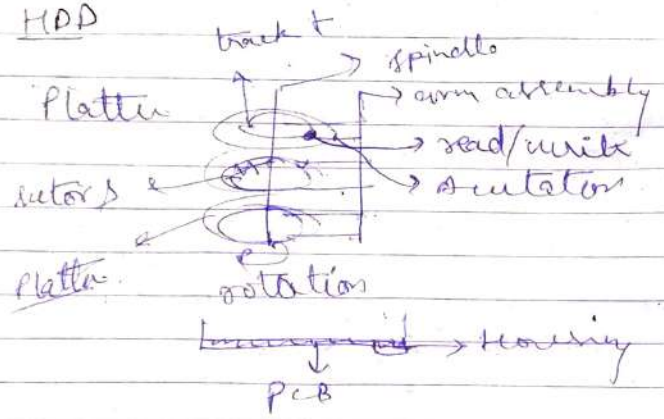
Types of comp storage

CPU ⟷ RAM
CPU: Cache
Cache → PRIMARY → RAM

HDD
secondary storage

CD-RW
USB
TAPE
offline storage

HDD



RPM

1 sector = 512 bytes

Seek time) time to track

(latency time) Track to sector

transfer rate) time taken in change of sector

SSD (semiconductor)

- flash memory
- Sata
- msata |2.5|
- M.2 sata (PCI-E)
- SSHD

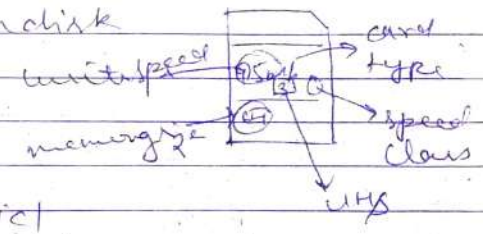
360 KB → 2.44 MB

floppy disk	80 data track	64 words	360 RPM

magnetic tape

Fried circuitboard

CD	DVD	Blu-ray	flash drive	memory card
700 mb	4.7 gb		2 gb – 128 gb	2 – 128 gb
↓ James Russel	↓ philips		↓ Sandisk	



Input devices

	QWERTY/ERGONOMIC
Key (108)	
mouse	
Joystick	
Lightpen	
Track ball → ball on mouse → touch pad	
scanner	
graphic tab → pen	
microphone.	

- SD
- camera
- mini SD
- micro SD
- mobile

MICR	magnetic ink character recognition
OCR	optical character recognition
Bar Code Reader	
OMR	optical mark reader
ATM	Automatic teller machine / use Thermal Printer (ROM)
Touch Screen	
fingerprint scanner	
Eye Scanner	

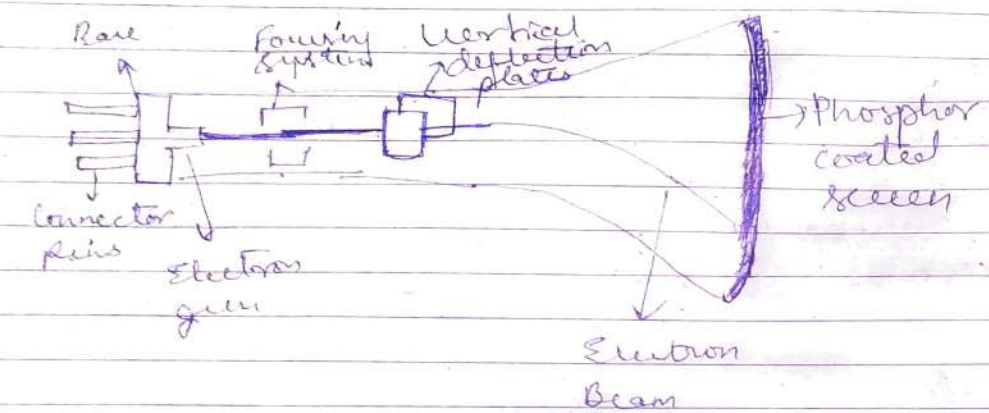
output devices

Projector	
Soundcard (inside laptop)	
Speaker	
Speech-generating device	
TV / video card	

output device

3D printer	
Braille embosser	
Braille reader	
flat panel	
GPS	
Headphones / Earphones	3.5 mm
computer output microfilm (COM)	
monitor / Karl F B (1897)	monochrome / gray-scale / color (16 / 256 / 65 / 16 / 16 / 16

CRT.

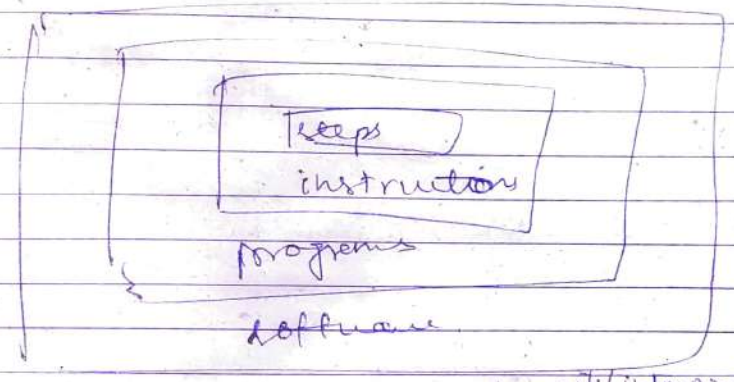


Video standard

	1981	
color graphics adapter	4 colors	320×200
Enhanced graphics adapter	16	
video graphics array	256	320×200
extended graphics array	16 lac	800×600
super video graphics array	1 crore lac	1280×

Pixel | Resolution | Refresh Rate | Dot Pitch | interlacing | Bit mapping | monitor size | Aspect ratio.

Software



compiler/interpreter/Assembler

- system → OS | Translator | linker | loader | debugger
- application
- utility → defrag | scan disk | anti virus.

System

324. OS
- process management
- memory "
- Disk "
- network
- security "
- Device Drivers.

translator

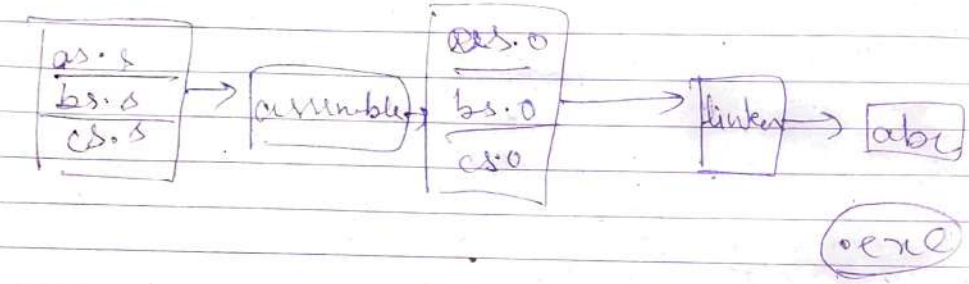
assembler — assembly → machine

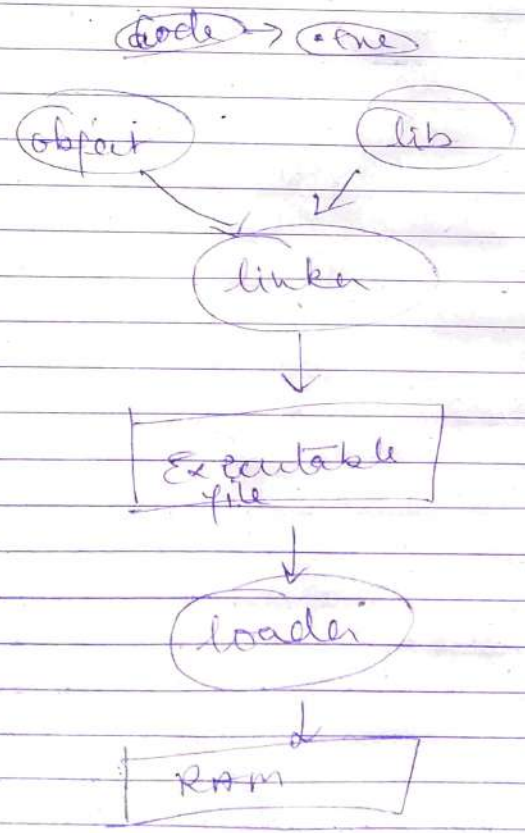
compile — source code → machine speed↑ error↓

interpreter — source code → execute speed↓ error↑

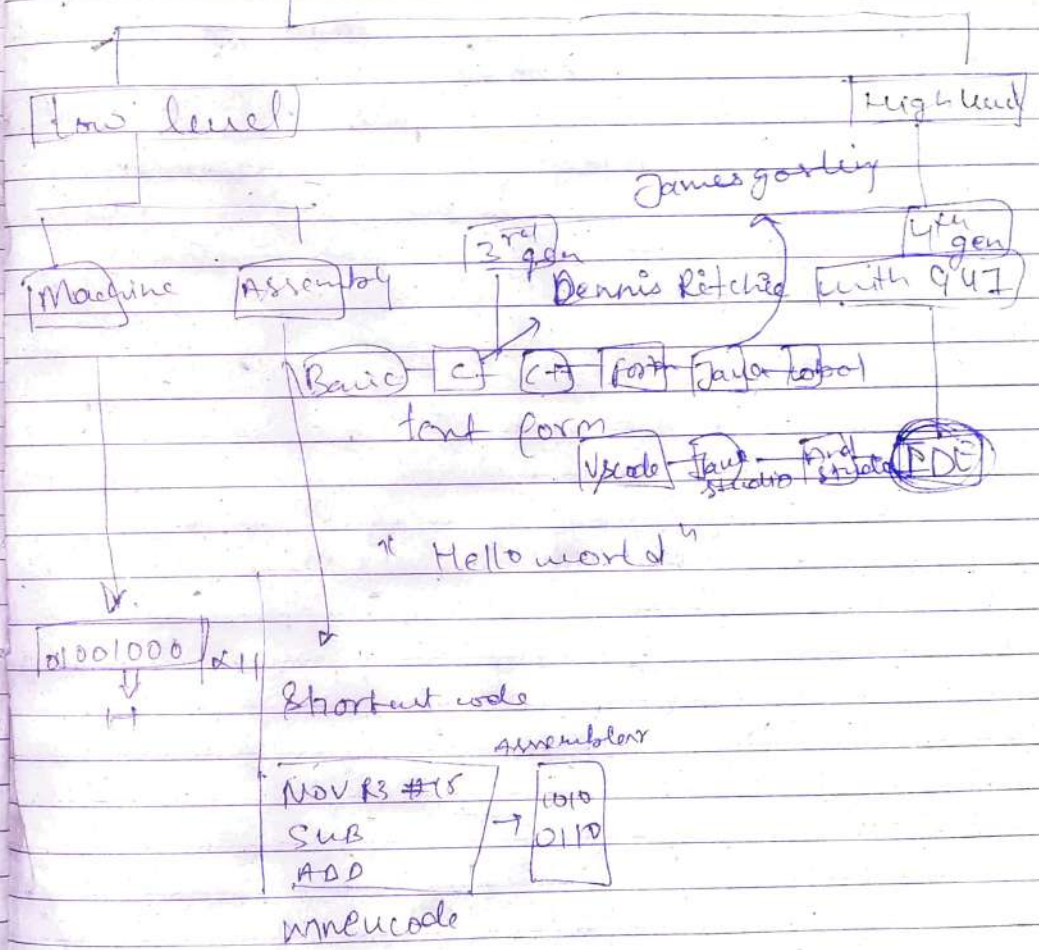
PAGE NO.
DATE: / /

Linker





Programming lang.



AI — PROLOG 1972 France
LISP 1959 John mccarthy

Communication
(Protocol)

Simplex →	Radio
⇄ one at a time Half duplex	walkie talkie
⇄ simultaneous full duplex	phone

USB

Serial | Parallel (8bit port)

time ↑ money ↓ (mouse/modem)	time ↓ money ↑

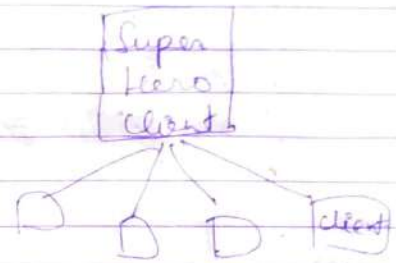
Cable

Twisted — unshield / shielded → pair cables → jacket

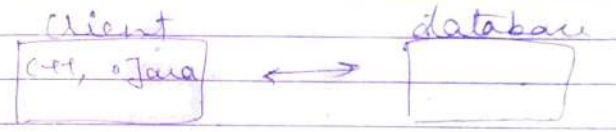
coaxial cable — bandwidth ↑ 10 Mbps
LAN
R-8/9/58/59.

optical fiber cable → jacket, cladding, core

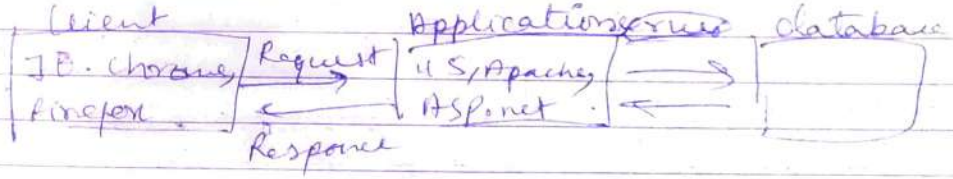
client server architecture



2-Tier



3-Tier



Internet

ISP → connection → modem → comp → brain
input ← website ← searchengine

Dialup connection	telephone wired
ISDN connection	Hello
Leased line connection	High speed Hello
VSAT connection	private network, fixed bandwidth, sat + router
Broadband connection	cables
wireless connection	Router (usb)
Usb modem connection	(sim card)

modem
modulation | de-modulation

Lan

wired (wireless, wired n/c)	wireless
→ NIC (or ceniar/ethernet card)	→ speed
→ cable connector	→ spectrum
→ hub/switch/router (TCP/IP)	
→ software	
→ power supply	
→ (patient at one) without break (data link layer)	
→ NOS / Bridge (two networks) / gateway (diff protocol, top layer)	

MAN (75km) (200 MB).

WAN (wireless)

VPN (Remote access client connections)

OSI (open standard interconnection) (data) layers

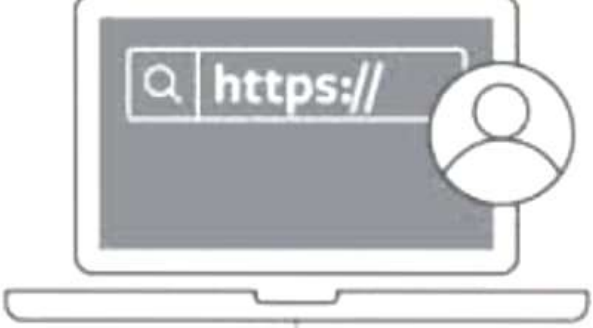
Application	software layers
Presentation	
Session	
Transport → Heart of OSI	
Network	Hardware layers.
Data link	
Physical	

sender ↓ receiver ↑

APS Tunnel which reduce pagal.

Topology

Bus | Ring (weakness) | mesh (2 NIC) | star | fully connected | line | tri



7 Layers of the OSI Model

7. Application

- End User layer
- HTTP, FTP, IRC, SSH, DNS

6. Presentation

- Syntax layer
- SSL, SSH, IMAP, FTP, MPEG, JPEG

5. Session

- Synch & send to port
- API's, Sockets, WinSock

4. Transport

- End-to-end connections
- TCP, UDP

3. Network

- Packets
- IP, ICMP, IPSec, IGMP

2. Data Link

- Frames
- Ethernet, PPP, Switch, Bridge

1. Physical

- Physical structure
- Coax, Fiber, Wireless, Hubs, Repeaters

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium